Sadeeq Adeyemi

Multi-object tracking and segmentation under occlusion

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RESEARCH INTERESTS

Multi-object tracking and segmentation (MOTS), occlusion reasoning, state estimation and data association, video foundation models (SAM 3), and evaluation on public benchmarks (KITTI-MOTS, MOTS20).

EDUCATION

MS, Data Science and Engineering, University of Maine, Orono, USA — 2025 – present

MS, Structural Engineering, University of Maine, Orono, USA — 2023 – 2025

MSc, Project Management, University of Portsmouth, United Kingdom — 2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria — 2019

RESEARCH EXPERIENCE

Graduate research: identity through occlusion in MOTS, University of Maine — 2026 – present

How much identity can a tracker keep while an object is hidden, and what earns it? Studied in simulation, on real driving geometry, and with Meta's SAM 3 and SAM 3.1 run zero-shot.

- Benchmarked SAM 3 and SAM 3.1 with the official HOTA evaluation. KITTI-MOTS val, cars: HOTA 82.6 and 74.6. MOTS20 train: 72.6 and 61.5.
- Defined an event-level measure from MOT17 visibility labels: SAM 3 kept identity through 94% of 245 occlusions, SAM 3.1 63%, with both falling for hides of 2 seconds or more.
- Showed from the source code and a controlled run that about half of SAM 3.1's deficit comes from its shipped tracking settings (car HOTA 74.0 to 78.3 against SAM 3's 82.3), and found a fault that stops it on long, crowded sequences.
- Built a Kalman-filter identity layer on SAM's output: it repaired SAM 3.1 (identity through occlusion 63% to 70%, McNemar $p = 0.004$) but not SAM 3. Motion outside the tracker says where an object is, not who it is.
- Mined 10,871 pedestrian occlusions in 1,256 NVIDIA driving clips with a line-of-sight test; on that geometry an occlusion-aware tracker kept 57% of identities through 1 to 5 second hides, against 2% for a standard tracker.
- Built a simulation lab for controlled occlusion experiments: identity kept rose from 0% to 89% as visibility reasoning, negative information and appearance re-identification were added; MathWorks JIPDA rose from 0% to 57% with the same visibility model.

Robotics Research Assistant, University of Maine Robotics Laboratory — 2025 – present

- Build mobile robots end to end, from sensor integration and wiring to ROS 2 nodes and lidar point-cloud perception.
- Implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

RESEARCH OUTPUTS

- Technical report: *Solving occlusion in MOTS: SAM 3 and SAM 3.1 zero-shot*, October 2026. Available on request.
- Code: SAM 3 / SAM 3.1 benchmark and evaluation, github.com/sadeeqgandalf/av-stack.
- Code and live lab: occlusion-aware tracking, github.com/sadeeqgandalf/lunar-occlusion-tracking.

INDUSTRY EXPERIENCE

Materials Engineer, R.W. Gillespie, Maine — 2024

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria — 2018 – 2023

- Deployed a computer vision model for real-time fatigue and defect detection in steel; automating the test cut its time by 50%.

TECHNICAL SKILLS

Methods: multi-object tracking, Kalman filtering (EKF, UKF), data association (Hungarian, JPDA, MHT), instance segmentation, re-identification

Software and data: Python, C++17, PyTorch, ROS 2, OpenCV, filterpy, MATLAB, TrackEval; KITTI, KITTI-MOTS, MOTS20, MOT17, NVIDIA PhysicalAI-AV

Certifications: AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect, Associate